MIL-F-80089B
30 September 1986
SUPERSEDING
MIL-F-80089A
5 June 1973

MILITARY SPECIFICATION

FURNACES, HEAT-TREATING, GAS-FIRED, BOX-TYPE, DIRECT-FIRED AND FULL-MUFFLE DESIGN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers horizontal box-type, gas-fired, heat-treating furnaces of the directly fired and full-muffle design with equipment and accessories specified herein.

1.2 Classification. The furnaces shall be of the following types, classes, and sizes. The type, class, and size to be furnished shall be as specified (see 6.2.1).

Type I - Direct fired
  Class 1 - 600 degrees F to 2000 degrees F operating temperatures.

  Class 2 - 1200 degrees F to 2500 degrees F operating temperatures.

Type II - Full muffle
  Class 1 - 1400 degrees F to 2000 degrees F operating temperature.

  Class 2 - 1800 degrees F to 2500 degrees F operating temperature.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1 Size. The inside working dimensions of the furnace should be selected from the size listed in table I for the type furnace required. Dimensions specified shall be minimum inside rectangular working dimensions. The length shall be the inside heated portion of the hearth, and shall not include the insulation thickness of the door opening.

TABLE I. Furnace size in inches.

Type I			Type II		
Width	Length	Height	Width	Length	Height
12	18	7	8	15	6
12	24	7	12	21	9
12	36	7	12	36	9
15	24	10	17	25	12
15	36	10	20	42	10
15	48	10	22	41	14
18	24	12			
18	36	12			
24	36	20			
24	48	24			

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-3296 - Forges, Furnaces, and Ovens (Exclusive of Space Heating and Cooking) Packaging of .

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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the nongovernment documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NEMA ICS 1 - Industrial Control and Systems.

ANSI/MC 96.1 - Temperature Measurement Thermocouples.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

FACTORY MUTUAL ENGINEERING CORPORATION (FM)

Handbook of Industrial Loss Prevention.

(Application for copies should be addressed to Factory Mutual Engineering Corporation, 1151 Boston - Providence Turnpike, Norwood, MA 02062.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. Furnaces and equipment provided in accordance with this specification shall be new and one of the manufacture's current models. The furnaces shall be of the oven-box design, having a hearth to support the charge and a single door located in the end of the furnace for inserting and removing the charge in a horizontal plane. Unless otherwise specified (see 6.2.1), the furnace shall be designed to operate on a fuel supply of 1000 British thermal unit (Btu) per cubic foot of natural gas at three- to six-inch water column pressure. Each furnace shall be furnished complete with the necessary burners, blowers, gas-air mixers, pressure regulators, valves, pilots, manifolds, vents, and piping so that when installed and connected to supply sources, it can be used in any operation required herein.

3.2.1 Type I. Type I furnaces shall be of the direct-fired design with the work directly exposed to the heat source. The combustion system shall be single-valve control, and capable of being adjusted to produce a furnace atmosphere that is either neutral, oxidizing, or reducing.

3.2.2 Type II. The type II furnaces shall be provided with a full muffle, and shall be designed and equipped for operation with a controlled atmosphere of the type specified (see 6.2.1).

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3.2.3 Measurement systems. Unless otherwise specified, either the U. S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the furnace. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the furnace that present safety hazards. The safety devices shall not interfere with the operation of the furnace. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the furnace, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the furnace itself. The safety systems of the furnace shall conform to the requirements of "FM Handbook of Industrial Loss Prevention." Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.5 Mercury restriction. The furnace shall not contain mercury or mercury components nor be exposed to free mercury during manufacture.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the furnace.

3.2.7 Environmental protection. The furnace shall be so designed and constructed that under the operating, service, transportation, and storage conditions described herein, the furnace shall no emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in affect at the point of installation.

3.2.8 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.9 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.2.10 Accessibility. All parts subject to wear, breakage or distortion and all parts which require periodic maintenance shall be accessible for adjustment and replacement, as applicable.

3.2.11 Material. All material used in the fabrication of the furnace and related equipment shall be of the quality necessary to produce a furnace to meet the requirements described herein. Materials that are exposed to the internal environment of the furnace chamber shall be compatible with the atmosphere and temperature conditions they will encounter.

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3.2.12 Reclaimed materials. The furnace may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the furnace.

3.3 Construction. The furnace shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the furnace to its maximum rating and capacity without distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the furnace shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the furnace and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.4 Electrical equipment. The furnace shall be furnished complete with all accessory electrical motors, contactors, controls, relays, transformers, signal lights and switches prewired where possible for operation upon connection to source of power. All power operated, manually adjusted and auto-

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matic controls shall operate on and shall be wired for 115 volts. The electrical system and enclosures shall conform to NEMA Standard ICS.1

3.4.1 Motors. Motors rated in excess of 1/4 horsepower shall be of the dual voltage, 230/460-volt, 60-hertz (Hz) type, and unless otherwise specified (see 6.2.1), designed and wired to operate on 460-volt, three-phase power supply. Motors shall conform to NEMA Standard MG 1.

3.4.2 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure or solder type. When soldered connections are used, the connections shall be mechanically secured before soldering. Rosin-base fluxes only shall be used in soldering operations. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness.

3.5 Performance. Each furnace shall be capable of operating continuously at its maximum rated temperature with a charge of not less than 80 pounds of iron or steel per square foot of hearth area. Minimum gas input and heat requirements shall be as specified in table II and III for the type, class and size furnace.

3.5.1 Temperature uniformity. With the furnace charge and the temperature stablized, the temperature variation from control point in the chamber to any other point in the chamber working area shall not vary more than $\pm 15^{\circ}$ F for type II, or $\pm 10^{\circ}$ F for type I furnaces. Each furnace shall be capable of maintaining the temperature uniformity specified with any size charge up to maximum load at any temperature control point within the operating temperature range of the furnace.

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TABLE II. Type I - furnace operating characteristics.

Size, furnace Working Dimensions in inches			Class 1		Class 2	
Width	Length	Height	Gas input 1000s BTU/ Hour	Hours to heat 70°F to 1600°F (Not more than)	Gas input 1000s BTU/ Hour	Hours to heat 70°F to 2300°F (Not more than)
12	18	7	205	.9	335	1.5
12	24	7	245	.9	400	1.5
12	36	7	310	.9	500	1.5
15	24	10	250	1.0	475	1.5
15	36	10	325	1.0	600	1.5
15	48	10	400	1.0	760	1.5
18	24	12	300	1.0	500	2.0
18	36	12	400	1.25	650	2.0
24	36	20	450	1.25	700	2.0
24	48	24	500	1.25	750	2.0

TABLE III. Type II - furnace operating characteristics.

Size, furnace Working Dimensions in inches			Class 1		Class 2	
Width	Length	Height	Gas input 1000s BTU/ Hour	Hours to heat 70°F to 1600°F (Not more than)	Gas input 1000s BTU/ Hour	Hours to heat 70°F to 2300°F (Not more than)
8	15	6	150	2.0	260	2.5
12	21	9	185	2.0	350	3.0
12	36	9	275	2.0	450	3.0
17	25	12	275	2.0	450	3.0
20	42	10	370	2.5	640	3.0
22	41	14	350	3.0	700	3.0

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3.6 Components.

3.6.1 Shell. The outer shell of the furnace shall be constructed of steel plate, not less than 3/16 inch in thickness, reinforced with steel members and welded throughout. End plates around door opening shall be so designed and of such material and construction that warpage will be prevented. Individual legs shall be braced and provided with bearing plates. Furnaces up to 24 inches shall have a work height above the floor of 40 inches, ±2 inches, and furnaces over 24 inches wide shall have a working height, above the floor, of 30 inches ±2 inches.

3.6.2 Hearth. Each furnace shall be provided with silicon carbide hearth plates at least 2 inches in thickness. If the type I furnace is of the under fired design, the hearth plates shall be provided with ledges on each side approximately 1/3 the height of the door opening. The hearth shall be capable of supporting a load in the heating chamber equal to at least 80 pounds per square foot of hearth area at maximum operating temperature, without permanent distortion. Hearth plates shall be capable of being removed and replaced with means provided for expansion and contraction. When specified (see 6.2.1), roller or slide rails shall be provided for moving the charge in and out of the heating chamber.

3.6.3 Insulation. The lining of the furnace shall be selected to minimize atmosphere contamination and resist abrasion and spalling, and shall consist of refractory or insulation brick, backed up with ceramic fiber, insulating brick, block or preformed insulation. Quality and thickness of the lining and back-up insulation shall be such that the outside surface temperature of the class 1 furnaces shall not exceed 100° F above ambient temperature after equilibrium has been established at maximum operating temperature. Powder, shredded, or granular material shall not be used for insulating purposes. When mortar is used, it should be of the same material as the units being cemented together.

3.6.4 Muffle. Type II furnaces shall be provided with a full silicon carbide muffle. The muffle shall be of gas-tight construction, and provided with atmosphere inlet connections and vents for operating with the type atmosphere specified in the contract or purchase order. Sufficient means shall be provided for expansion and contraction, and the muffle shall be capable of being removed and replaced with minimum disturbance to the insulation.

3.6.5 Door. Unless otherwise specified, the furnace door shall be of the manually operated, vertical-lift type, with counter weights. When required (see 6.2.1), the door shall be power operated and provided with a foot control valve. The door shall be fully insulated with ceramic fiber, refractory or insulating material, and shall overlap the brick facing of the furnace at least 2-1/2 inches on all four sides. Each door shall be provided with jambs and clamps for tight closure. The door on the type I furnace shall seal brick to brick. Type II furnace shall be provided with gaskets to form a gas-tight unit. The door opening dimensions shall be equal to the

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width and height of the furnace working dimensions. The door shall be provided with a nonfogging sight glass for observing the charge without opening the door.

3.6.6 Heating system. Each furnace shall be provided with all necessary burners, gas-air mixing systems, governors, manifolds, valves, blowers, gauges, pilots, vents, and safety devices required for safe, continuous operating. Burners shall be sufficient in number and size to meet the heating requirements and temperatures uniformity specified herein. Burners shall be arranged or located in such a manner that the flame will not impinge on the charge or muffle.

3.6.6.1 Recuperator. When specified (see 6.2.1), heat recuperator system shall be provided to recovered the heated gases and improve the efficiency of the burners.

3.6.7 Thermocouple. Each furnace shall be equipped with thermocouple assemblies complete with mounting heads, insulators, and atmosphere protective tubes. One thermocouple shall be used with the indicating, recording, and controlling instrument, and one with the excess temperature control instrument. Furnaces having a temperature rating in excess of 2000° F shall be furnished with platinum-rhodium-type thermocouples. Furnaces having a temperature rating of 2000° F and lower shall be furnished with chromel-alumel type thermocouples. The thermocouples shall conform to ANSI/MC96-1. The thermocouple lead wires shall be at least 25 feet in length, and of the same material as the thermocouple or of a material having the same thermoelectric characteristics at a thermocouple head temperature within 0° to 250° F.

3.6.8 Multi-zone control. When necessary to maintain the temperature uniformity and performance requirements specified for the furnace, the chamber shall be divided into a suitable number of control zones. The controls and instruments, conforming to the requirements specified herein, shall be provided for multi-zone control.

3.6.9 Instruments. Each furnace shall be provided with a single, self-contained temperature indicating-recording-controller type instrument, with an accuracy of 0.50 percent of full scale reading. Solid-state design shall be employed where possible. Instrument operating characteristics and dimensions shall conform to the requirements of 3.6.9.1 through 3.6.9.4. The instrument shall be marked to indicate the particular type of thermocouple for which it is calibrated. Automatic reference junction compensation shall be provided in all instruments, using thermocouples as the primary temperature sensing element.

3.6.9.1 Microprocessor. When specified (see 6.2.1), the furnace control system shall incorporate a microprocessor digital programmable/controller to provide maximum control flexibility within the limitations of the furnace's designed capability. The microprocessor shall provide for not less than 50 read/write programmable segments of random access memory and input into the system shall be by manual data input (MDI), manual keyboard and magnetic tape cassette.

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3.6.9.2 Temperature recorder. The temperature recorder shall be of the strip chart, self-balancing type. Unless otherwise specified, the calibrated chart width shall be not less than 6 inches, and the chart length shall be not less than 80 feet to the roll. The chart speed shall be one inch per hour. The chart mechanism shall be furnished in a dust-proof housing with a glass door that permits full view of chart length. When required, the chart width shall be at least 9 or 12 inches as specified (see 6.2.1).

3.6.9.3 Temperature controller. The temperature controller shall be of the proportioning, automatic, controlling type. The controller shall be capable of maintaining the set control temperature with a maximum deviation of not more than $\pm 2^{\circ}$ F. The control system shall provide thermocouple up-scale fail-safe protection that will shut off the heat source when the furnace temperature exceeds approximately 100° F above maximum.

3.6.9.4 Temperature indicator. The control indicator shall be approximately the same length as the chart width and have a maximum range of not less than 100° F above the maximum operating temperature of the furnace. The increments between successive temperature marks shall be not greater than 20° F, with the spacing of numbered marks not greater than 400° F.

3.6.9.5 Excess temperature control. The furnace shall be provided with an excess temperature control circuit independent of the control circuitry. The controller shall function to shut off the heat source to the furnace when the over-temperature control point is reached. In case the furnace is shut down by the excess temperature control system, the system shall sound an audible alarm, and the furnace shall remain off until started manually. Calibrated accuracy shall be within one percent of full-scale reading.

3.6.9.6 Instrument panel. Floor mounted instrument panels conforming to NEMA ICS-1 shall be provided for mounting all instruments and accessory controls. All ventilated openings shall be designed to prevent entrance of any deleterious substance. Signal lights shall be provided to indicate the furnace cycle performance. The panel shall provided flush or semi-flush mounting of all instruments.

3.6.9.7 Thermocouple burn-out alarm signal. The excess temperature cutout and the temperature controller shall be designed so that loss of either thermocouple circuits will provide an audible and visible alarm signal.

3.7 Atmosphere generator. When specified (see 6.2.1), the furnace shall be furnished with a protective atmosphere generator, complete with all necessary piping, control, and safety equipment. Type of atmosphere, size of generator, cubic feet per hour, carbon content, and dew point requirements shall be specified.

3.8 Flame curtain. When specified (see 6.2.1), a gas-air-flame curtain shall be provided for the door opening, and shall operate automatically by limit switches through solenoid valves to protect the atmosphere when the door is open.

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3.9 Loading mechanisms. When specified (see 6.2.1), loading and load extracting equipment shall be furnished. Design and requirements shall be as specified.

3.10 Shelves. When specified (see 6.2.1), the type I furnace shall be provided with removable open-grid type shelves. Number and arrangement of shelves shall be as specified.

3.11 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.12 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each furnace and contain the information listed below. If the furnace is a special model, the model designation shall include the model of the basic standard furnace and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.13 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.14 Workmanship. Workmanship of the furnace shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article. When first article approval is required under 3.1, first article inspection shall be performed. Unless other specified (see 6.2.1), first article inspection shall comprise the examination in 4.6 and all tests in 4.7. Failure of the item to pass any examination or test shall be cause for disapproval of the first article.

4.5 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7.1, 4.7.2, and the inspection in 4.8. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

4.6 Examination. The furnace and equipment shall be examined for design, dimensions, construction, materials, components, electrical and safety compliance, and workmanship to determine compliance with the requirements of this specification.

4.7 Tests. All tests shall be conducted at an ambient temperature of 77° F ±18° F. Test instruments and sensing elements used to perform the test specified for the furnace and instruments shall have been checked against a standard potentiometer type instrument and calibrated thermocouple of known accuracy within the previous six months to read accurate within ±2° F.

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4.7.1 Instrument test. Indicating and recording instruments shall be tested by impressing various predetermined voltages, simulating thermocouple outputs for different furnace temperatures to terminals where the thermocouples would normally be connected. These tests shall be made in accordance with, and results checked against, standards established ANSI/MC 96.1. The instruments shall be tested through their operating range.

4.7.2 Circuit test. Each circuit of the electrical system shall be tested for dielectric strength, continuity, insulation-resistance, and faulty grounds in accordance with the requirements of ANSI/NEMA ICS 1.

4.7.3 Performance test. The furnace shall be prepared for operating, and operated in accordance with the manufacturer's handbook of instructions until equilibrium has been established at maximum operating temperature. After equilibrium has been established and the refractory and insulation has been cured, the furnace shall be allowed to cool to a work chamber temperature of 77° F ±18° F. The regulating controls shall be adjusted to attain maximum operating temperature. The furnace shall heat to at least the temperature specified, within the time limit specified for the type, class, and size furnace under test. the furnace shall continue in operation until equilibrium is again established at maximum operating temperature. The surface temperature shall no exceed the maximum surface temperature specified in 3.6.3 for the class furnace under test. Each safety device shall be tested by simulating a failure that would normally cause the devices to function. Each device shall perform its intended function.

4.7.4 Temperature uniformity test. The furnace shall be loaded with a production type charge equivalent to at least 80 percent of maximum load, and a chamber temperature survey performed. A minimum of nine test locations shall be used, with one thermocouple located in each corner of the chamber and one thermocouple located within six inches of each control thermocouple. At least three survey readings shall be taken at intervals of five to ten minutes with equilibrium established at minimum and maximum operating temperature. The maximum temperature variation between the control point in the chamber and all points under test shall not exceed the temperature uniformity requirements specified in 3.5.1 for the type furnace under test.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), preservation, packing, and marking shall be accomplished in accordance with MIL-F-3296 as specified.

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6. NOTES

6.1 Intended use.

6.1.1 Type I. The type I furnace is intended primarily for use in shops for hardening, tempering, normalizing, and annealing metal parts with a limited degree of atmosphere control obtained from the firing system.

6.1.2 Type II. Type II furnace is intended for most heat-treatment processes that require the work to be protected from the products of combustion. The furnace may be used with or without a manufactured atmosphere.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.

b. Type, class, and size (width, length, height) required, see table I and 1.2).

c. First article approval, if required (see 3.1).

d. Type of fuel, if different (see 3.2).

e. If furnace is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.3).

f. Type of manufactured atmosphere for which furnace will be used (see 3.2.2). Type II furnace only.

g. Specify, exceptions and safety and health requirement (see 3.2.4 and 6.4).

h. Painting if different (see 3.3.5).

i. Motor voltage and phase, if different (see 3.4.1).

j. Hearth slide or roller rails, if required (see 3.6.2).

k. Power operated door, if required (see 3.6.5).

l. Recuperator, if required (see 3.6.6.1).

m. Microprocessor, if required (see 3.6.9.1).

n. Temperature recorder chart width, if different (see 3.6.9.2).

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o. Atmosphere generator, if required (see 3.7).

p. Flame curtain, if required (see 3.8).

q. Load mechanism, if required (see 3.9).

r. Shelves, if required (see 3.10).

s. List of repair parts, if required (see 3.11).

t. Nameplate, if different (see 3.12).

u. First article inspection, if different (see 4.4).

v. Quality conformance inspection, if different (see 4.5).

w. Specify level of preservation, and packing required, if different (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Subject term (key word) listing.

Heat treating
Gas fired
Box type
Direct fired
Full muffle

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activities:
Navy - MC, OS

Preparing activity:
DLA - IP

Project (3424-0109)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-80089B	Furnaces, Heat-treating, Gas-fired, Box-type, Direct-fired and Full-muffle Design

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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